The purpose of this document is to provide steps to run EDGIE.

EDGIE is a MATLAB based tool to analyze the impacts of electric vehicles, heat pumps, and heat-pump water heaters on the distribution grid. It also includes a vehicle-to-home strategy to reduce the peak demand.

The authors used MATLAB 2022, CVX 2.2 , Tableau 2023.3 and macOS with M2/16 GB RAM to generate and plot the results.

This document will walk you through to install the required software and run the scripts.

The next page will include steps for downloading the required input files and running the EDGIE code. It will require 35-40 GB of space.

Software needed:

MATLAB

Download link: https://www.mathworks.com/help/install/ug/install-products-with-internet-connection.html

Downloading time will vary depending on the internet speed.

Step 1: Create a folder named “ComStock”

Step 2: Go to the link : 10.5281/zenodo.21628313

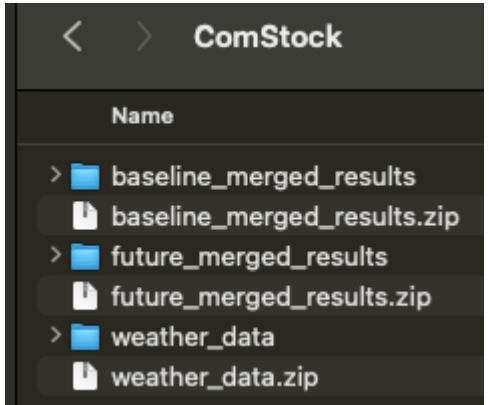
Step 3: Download the three zip files



Step 4: Move the downloaded files to “ComStock” folder

Step 5: Extract all the files

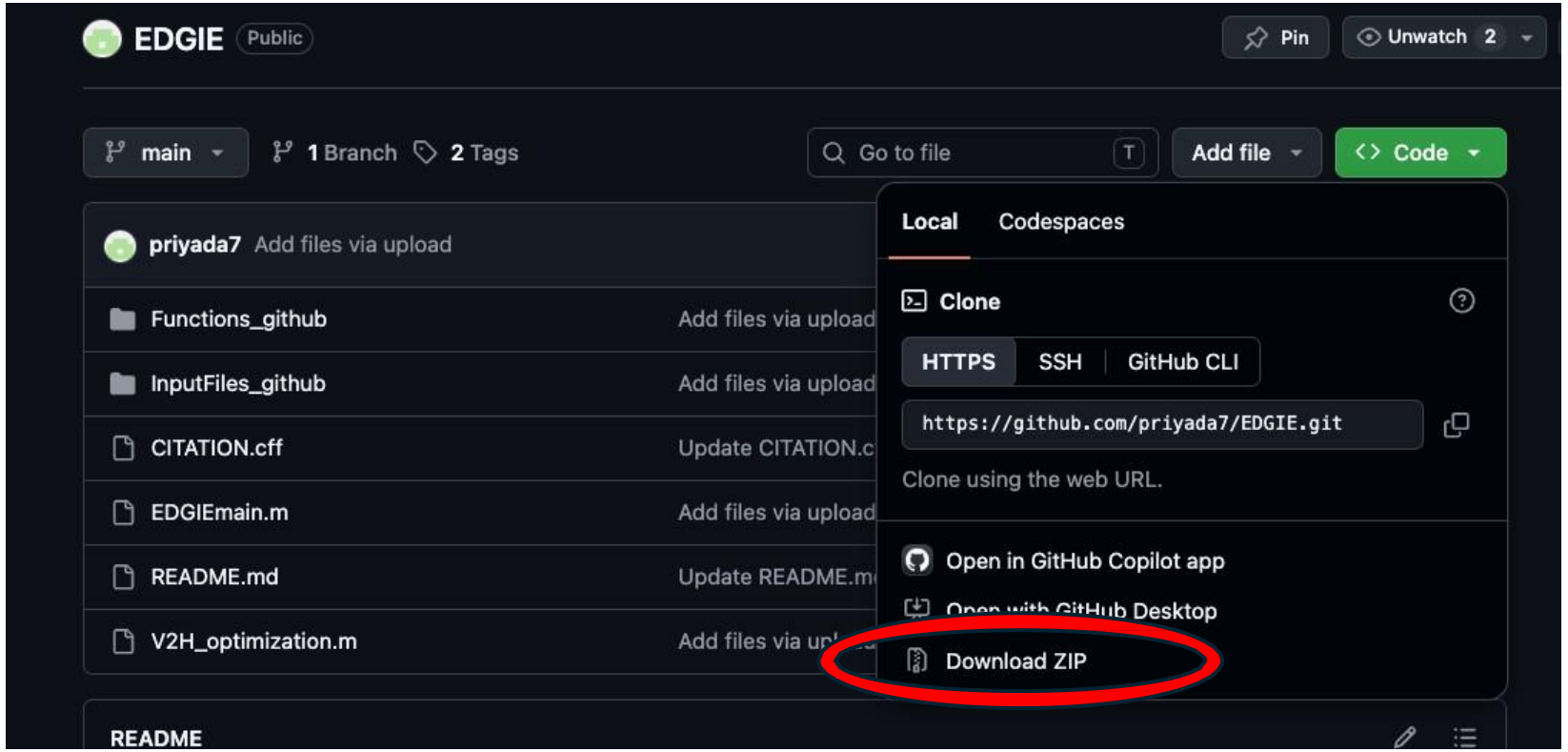
Step 6: ComStock folder should look like this



Step 7: Go to the github repository:

https://github.com/priyada7/EDGIE

Step 8: Click on code and then select download zip



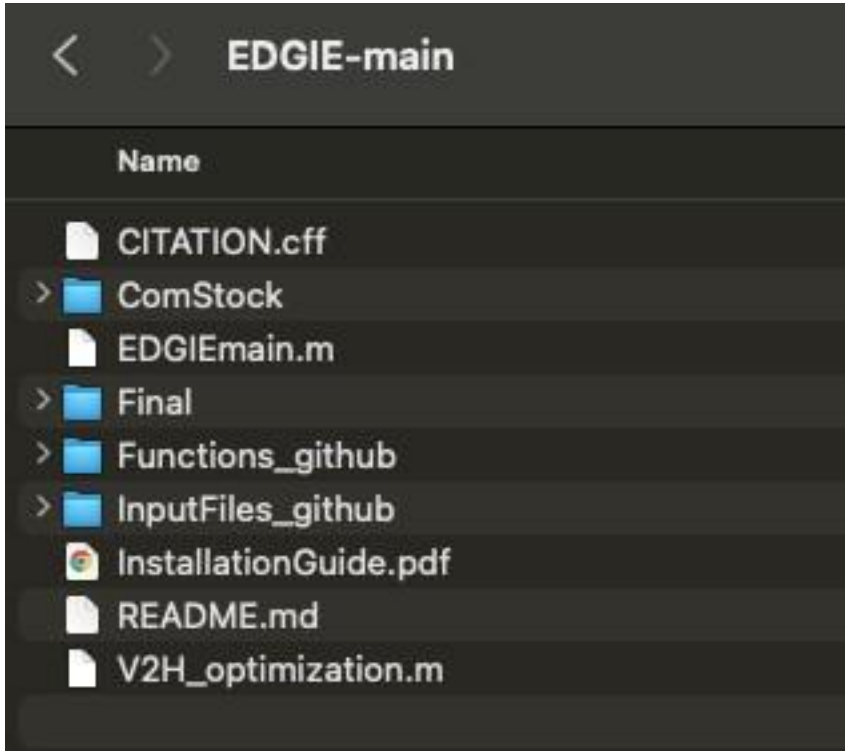
Step 9: Create a new folder “EDGIE Github”

Step 10: Move the zip folder EDGIE-main.zip in that folder

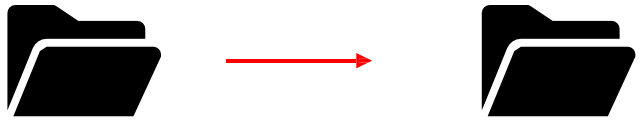
Step 11: Extract the zip file

Step 12: Move the ComStock folder to “EDGIE-main” folder

Step 13: The “EDGIE-main” should look like this

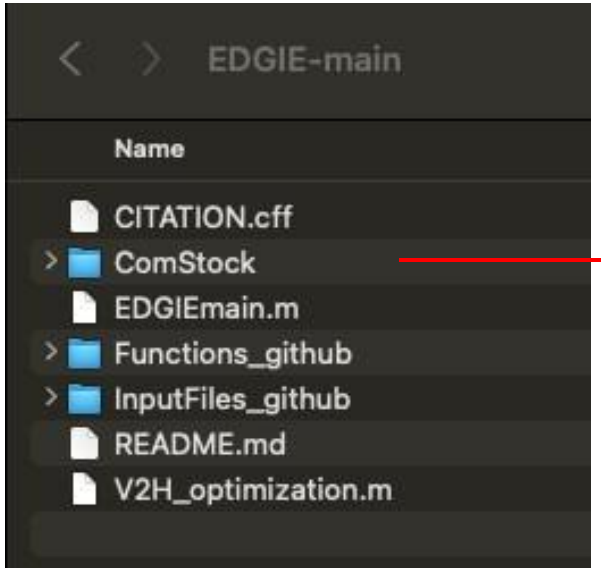


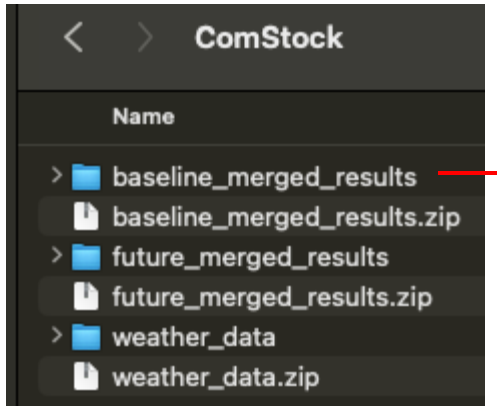
Lets re-verify the extracting process. Your folders should look like this



EDGIE Github

EDGIE-main







Step 14: Open EDGIEmain.m file

Step 15: Replace XXX with the working directory location (line 2- 5)

```
addpath 'XXX/EDGIE Github/EDGIE-main/Functions_github'
addpath 'XXX/EDGIE Github/EDGIE-main/InputFiles_github'

baseDir = 'XXX/EDGIE Github/EDGIE-main/ComStock';
```

Example : for macOS

```
addpath '/Users/priyadarshan/Library/CloudStorage/Box-Box/Herricks/Codes/EV/v2h/v2h new/untitled folder/EDGIE Github/EDGIE-main/Functions_github'
addpath '/Users/priyadarshan/Library/CloudStorage/Box-Box/Herricks/Codes/EV/v2h/v2h new/untitled folder/EDGIE Github/EDGIE-main/InputFiles_github'

baseDir = '/Users/priyadarshan/Library/CloudStorage/Box-Box/Herricks/Codes/EV/v2h/v2h new/untitled folder/EDGIE Github/EDGIE-main/ComStock';
```

Note: If using Windows use replace “/” with “\”

Step 16: Click Run

By default it will run for Minnesota and Hennepin county (stateID = 27) and output in command window should be of the order 5e9.

Note: The exact number may vary depending on the random numbers generated, version of MATLAB. Run time will be within a minute.

To run statewide location, change line 171 to

```
for stateIdx = 1:length(cities)
```

To run the V2H code,

It will first require installing CVX 2.2.

CVX can be downloaded from : https://cvxr.com/cvx/

The website also has steps to install the CVX

After installation, open the file V2H_optimization.m and specify the working directory and click run

Note: optimization will be slow, it is advised to use **MOSEK** as solver. One location with 100 home simulations should be complete in 40-50 seconds with MOSEK.

Software download links:

MATLAB

https://www.mathworks.com/help/install/ug/install-products-with-internet-connection.html

CVX

https://cvxr.com/cvx/

Plotting results

https://www.tableau.com/

In case of any issue, please reach out at priyada@purdue.edu